Question 1

Write a Python program that outputs the message `"Welcome to Computer Science!"` to the screen.

Answer:

```
print("Welcome to Computer Science!")
```

Question 2:

Write a program that prompts the user to enter their name, stores it in a variable called `student_name`, and prints a personalized greeting such as `"Hello, Alice!"`.

Answer:

```
student_name = input("Enter your name: ")
print("Hello, " + student_name + "!")
```

Question 3:

Write a program that asks the user to enter the width and height of a rectangle as whole numbers (integers). Calculate and display the total area using the formula:

$$area = width \times height$$

Answer:

```
width = int(input("Enter width: "))
height = int(input("Enter height: "))

area = width * height
print("Area:", area)
```

Question 4:

In computer science, finding remainders is a standard operation. Write a program that asks for an integer number of sweets and divides them equally between 4 children. Display both:

- How many whole sweets each child receives.
- How many leftover sweets remain.

Answer:

```
sweets = int(input("Enter number of sweets: "))

per_child = sweets // 4
remainder = sweets % 4

print("Each child gets:", per_child)
print("Sweets left over:", remainder)
```

Question 5:

Write a program that asks the user for their score out of 100. If the score is 50 or higher, the program should print `"Passed"`.

Answer:

```
score = int(input("Enter your exam score: "))

if score >= 50:
    print("Passed")
```

Question 6:

Write a program that prompts the user to enter an integer. Determine whether the number is **even** or **odd**, and print the result.

Answer:

```
number = int(input("Enter a whole number: "))

if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Question 7:

Write a program that asks the user for a secret password. If the input matches `"IGCSE2026"`, print `"Access Granted"`. Otherwise, print `"Access Denied"`.

Answer:

```
password = input("Enter password: ")

if password == "IGCSE2026":
    print("Access Granted")
else:
    print("Access Denied")
```

Question 8:

Write a program that takes an exam percentage (0 to 100) from the user and outputs the corresponding letter grade using this scheme:

- 80 and above: `"A"`
- 60 to 79: `"B"`
- 40 to 59: `"C"`
- Below 40: `"U"`

Answer:

```
mark = int(input("Enter mark: "))

if mark >= 80:
    print("Grade: A")
elif mark >= 60:
    print("Grade: B")
elif mark >= 40:
    print("Grade: C")
else:
    print("Grade: U")
```

Question 9:

A theme park ride allows riders who are at least 120 cm tall **and** at least 10 years old. Write a program that inputs height in centimeters and age in years. Output `"Allowed to ride"` if both conditions are met, otherwise output `"Not allowed to ride"`.

Answer:

```
height = int(input("Enter height in cm: "))
age = int(input("Enter age in years: "))

if height >= 120 and age >= 10:
    print("Allowed to ride")
else:
    print("Not allowed to ride")
```

Question 10:

Write a validation program that first checks whether an entered temperature in Celsius is within a realistic ambient range (between -20°C and 50°C inclusive).

- If the temperature is **outside** this range, print `"Invalid temperature"`.
- If it is **inside** the range, check if the temperature is 30°C or above. If so, print `"Hot"`; otherwise, print `"Moderate or Cold"`.

Answer:

```
temp = float(input("Enter ambient temperature in Celsius: "))

if temp < -20 or temp > 50:
    print("Invalid temperature")
else:
    if temp >= 30:
        print("Hot")
    else:
        print("Moderate or Cold")
```